ANALYZING STUDENT INTERACTION DATA WITH ARTIFICIAL INTELLIGENCE USING PYTHON

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The trainer

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the introduction

In light of the rapid evolution of AI tools, it has become clear that these technologies play an increasing role in supporting the educational process. This study aims to understand how students from different academic levels (High School, Undergraduate) interact with AI assistants like ChatGPT. By analyzing data representing 10,000 student sessions generated in a realistic synthetic way, we focused on student behavior, the types of tasks they requested help with, their satisfaction levels, and whether they returned to use the tool again. This work also highlights the opportunity to extract insights that can help develop more effective educational tools.

Data description and analysis objectives

TThis dataset covers student behavior when using AI tools for various academic tasks. The file contains 10,000 rows and 11 columns representing different characteristics of students and their interactions: StudentLevel, Discipline, TaskType, UsedAgain, SessionLengthMin, TotalPrompts, SatisfactionRating, RepeatCount, TimeStamp, StudentID, Notes.

We aim to provide a comprehensive view of how students use technology and analyze interaction patterns and preferences, focusing on educational and behavioral aspects.

Data Source;
This data was obtained from the Kaggle platform, from a dataset titled: "AI Assistant Usage in Student Life – Synthetic Dataset"

Objectives:

- Analyze student satisfaction with the use of AI tools.
- Study the frequency of tool usage among students.
- Present the relationships between variables using clear visual representations.
- Identify the factors that affect student satisfaction to improve their experience.

Data cleaning & Preprocessing

To ensure data quality and accuracy, the following preprocessing steps were performed:

1. Handling Duplicates:
   - Identified 45 duplicate rows within the dataset.
   - We Removed these duplicates to prevent any negative impact on the final analysis and results.

2. Managing Missing Values:
   - Certain columns contained a limited number of missing values.
   - Instead of deletion, missing data was handled through imputation:
     * Categorical Columns: Imputed with a default value of "**Unknown**".
     * Numerical Columns: Imputed using the Mean.

This approach was chosen to preserve the maximum amount of data and ensure comprehensive analysis.

3. Resolving Inconsistent Values:
   - The **TaskType** column contained inconsistent naming conventions, such as variations in spelling and letter casing.
   - We Standardized all values into a uniform format to eliminate noise and ensure consistency in the final results.

Data Cleaning & processing

4- Handling Outliers:

- We noticed some extreme values in the TotalPrompts column that didn't align with the rest of the data.
- After analyzing the distribution, we found that the realistic usage typically ranges between 1 and 5.
- We decided to remove any values outside this range to ensure our statistical results remain accurate and unbiased.

5- Data Type Conversion:

- We discovered that one of the numerical columns was incorrectly stored as a string (text) format.
- We converted this column to its correct numerical type (Integer/Float).
- This step was essential so we could perform mathematical operations and include it in our final calculations.

6- Final Review:

- Once we finished the cleaning process, we conducted a comprehensive audit to ensure there were no remaining errors, duplicates, or missing values.
- We confirmed that all columns are now fully validated and ready for advanced modeling and analysis.

Digital and statistical analysis

Our dataset consists of 10,000 rows and 11 columns. During the analysis phase, we conducted several statistical steps to ensure the accuracy and depth of our insights:

1-Analyzing AI Task Types:

- We performed a statistical analysis to identify the primary tasks where students utilized AI tools. and the result showed:

array(['studying', 'coding', 'writing', 'brainstorming', 'homework help','research', 'Unknown']

2- User Distribution Across Academic Levels:

- As part of our statistical breakdown, we analyzed the number of students using AI tools at each academic level.
- This helped us understand which student groups are most engaged with AI technologies in their learning process.

StudentLevel	Total
Undergraduate	5981
High School	2027
Graduate	1996
Unknown	2

3- Analyzing User Retention (Non-Returning Users):

- We specifically looked into students who completed their tasks using AI but chose not to use the tools again.
- Our analysis revealed that 924 students fall into this category, providing us with a clear metric on user retention and satisfaction.

Digital and statistical analysis

4- Session Duration Analysis:

- We calculated the average, minimum, and maximum session durations to understand user engagement patterns.
- Our findings showed that session lengths varied significantly, starting from a minimum of 0.03 minutes up to a maximum of 110.81 minutes.
- We determined that the overall average session duration was approximately 19.85 minutes.

5- User Satisfaction Ratings:

- As part of our statistical evaluation, we analyzed user satisfaction scores to measure the effectiveness of the AI tools.
- The ratings ranged from a minimum of 1.0 to a perfect score of 5.0.
- We found that the average satisfaction rating across all interactions was 3.41, indicating a generally positive reception.

6- A comparison of students who used artificial intelligence again and their evaluation of it was more than 4

- We compared the number of students who used AI again using the AND tool and rated it above 4, and the results showed that there were 2384 students.

7- Analysis of the number of dissatisfied students who did not complete their tasks

- We analyzed the number of students who were dissatisfied with the use of AI tools and gave up before completing the tasks; the number was 21 students.

Digital and statistical analysis

8- Analysis of Long Sessions (Over 1 Hour):

- We calculated the number of sessions that exceeded one hour to identify "power users" or complex tasks.
- Our analysis showed that 162 sessions lasted longer than an hour out of the total 10,005 sessions.

9- Low Satisfaction Rates in Coding Tasks:

- We specifically analyzed user satisfaction within the Coding category to identify potential challenges.
- We found that 643 students out of the 1,949 who used AI for coding reported low satisfaction levels, highlighting a need for better AI support in technical fields.

10- Comparison between students in Engineering OR Business

We used the OR tool to search for students majoring in Engineering or Business, and the results showed that there were 2841 students.

11- Comparison between Computer Science students and undergraduate students

We used the AND tool to search for students majoring in Computer Science at the undergraduate level, and the results showed that there were 868 students.

Digital and statistical analysis

12- Calculating the average number of orders per task using artificial intelligence

Using statistical analysis, we calculated the average number of commands sent to accomplish tasks using artificial intelligence, and the results showed the average number of commands for each task:

TaskType	TotalPrompts
Unknown	4.000000
brainstorming	6.271008
coding	5.527803
homework help	5.695053
research	5.714286
studying	5.530170
writing	5.536924

13- Calculate the number of times each type of task was used

Using statistical analysis, we calculated the number of times each type of task was used, and the result was:

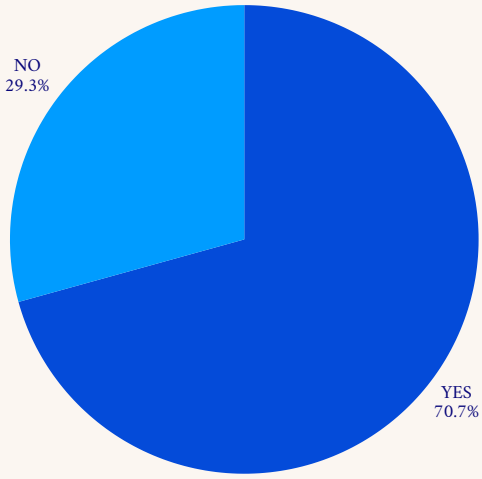
TaskType	Count
writing	3102
studying	2041
homework help	1959
coding	1949
brainstorming	476
research	476
Unknown	2

Visual Analysis

1- User Retention Rate (Repeat Usage)

Observations:

- 70.7% of students returned to use the AI tool, reflecting a high overall satisfaction and trust in the technology.
- 29.3% of users did not return, which indicates specific challenges or barriers that may have hindered their experience.
- **Strategic Insight:** There is a significant opportunity to increase user loyalty by investigating the reasons behind the 29.3% non-return rate and addressing potential pain points.

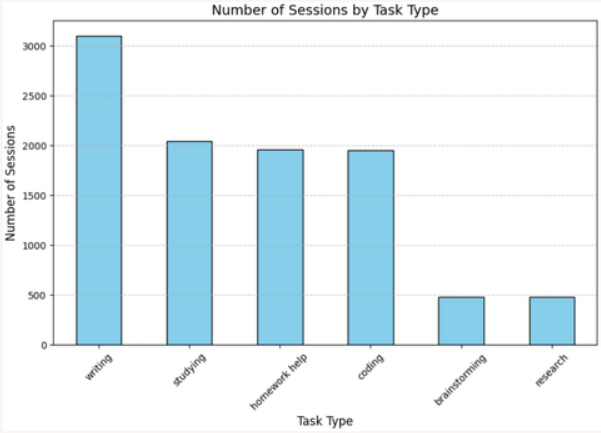


This chart illustrates the distribution of students who chose to use artificial intelligence again after their initial experience. The data indicates that 70.7% of students reused it, while 29.3% did not.

2- Distribution of Sessions by Task Type:

Observations:

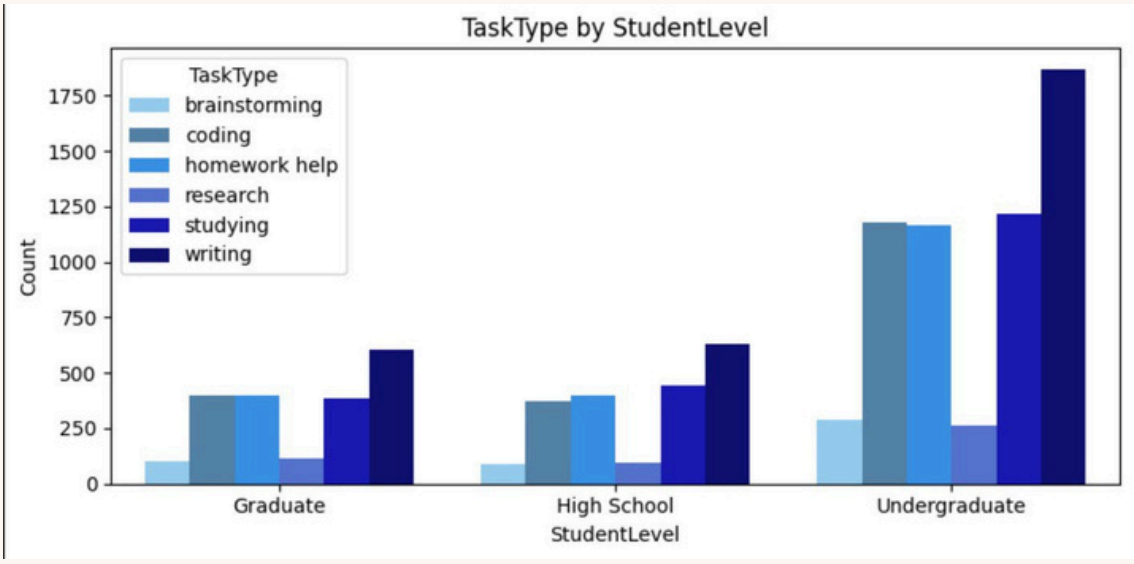
- Writing emerged as the most dominant task, showing that students rely heavily on AI for content creation and drafting.
- Research and Brainstorming were identified as the least frequent tasks in our dataset.
- **Strategic Insight:** We can enhance the overall utility of the tool by promoting and improving features related to research and brainstorming, encouraging students to explore the full potential of AI beyond just writing and studying.



This visualization breaks down the total number of sessions across different categories, including Writing, Studying, Coding, Research, Brainstorming, and Homework Help.

Visual Analysis

3- Task Distribution by Academic Level:



This chart displays the distribution of different Task Types across three main categories: Graduate, High School, and Undergraduate.

Observations

• Writing is the most demanded task across all academic levels, regardless of the student's Level.
•Undergraduate students are the most reliant on AI assistants compared to other groups.
• Tasks that require independent thinking, such as research or brainstorming, are not in high demand.

Results and interpretations:

- High overall usage rate:

- Result: Approximately 70.7% of students use AI tools repeatedly.
- Interpretation: This reflects high student satisfaction and indicates that these tools are effectively integrated into their academic workflows to improve performance.

- Student Levels Analysis:

- Result: Undergraduate students represent the majority of users, followed by High School and Graduate students.
- Interpretation: This highlights a heavy reliance on AI for undergraduate studies, while suggesting that High School students may still need more guidance or awareness.

- Distribution by Task Type:

- What we found: Writing is the most common task, followed by Studying and Coding.
- Our Interpretation: We noticed a clear need for student support in content creation and solving technical programming exercises.

- Least Common Tasks:

- What we found: Research and Brainstorming are the least used tasks.
- Our Interpretation: We believe this could be due to limited promotion of these specific features or because their benefits aren't fully clear to users yet.

Recommendations

- Raising Student Awareness:
We recommend that schools and universities launch awareness campaigns about AI tools and their benefits, specifically targeting High School students who show lower engagement rates.

- Integrating AI into Educational Activities:
AI tools should be incorporated into lessons and assignments. This will help students learn how to use these technologies correctly, effectively, and ethically.

- Providing Licensed Tools for Students:
Educational institutions should consider providing free or subsidized licenses for reliable AI tools (like ChatGPT Plus) to ensure all students have equal access to advanced features.

- Enhancing Underused Features:
Efforts should be made to demonstrate the value of AI in Research and Brainstorming. Highlighting these capabilities can help students move beyond basic tasks and utilize AI for more creative and analytical work.

References

- https://www.kaggle.com/datasets/ayeshasal89/ai-assistant-usage-in-student-life-synthetic